

Mohammad Reza Daneshvar Garmroodi

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EDUCATION

Ph.D., Concordia University, Montreal, Canada

Sep 2019 - May 2025

Advisor: Ida Karimfazli

Department of Mechanical, Industrial and Aerospace Engineering

Thesis title: Mixing Dynamics in Yield Stress Fluids

Committee: Rob Poole, Ali Dolatabadi, Alex De Visscher, Charles Basenga Kiyanda

M.A.Sc., Amirkabir University of Technology, Tehran, Iran

Sep 2017 - Aug 2019

Advisor: Ali Ahmadpour

Department of Mechanical Engineering

Thesis title: Numerical Study of Two-Phase Yield Stress Fluids and Water in Different Flow Regimes

Committee: Ehsan Amani, Mohammad Reza Hajmohammadi

B.Sc., University of Tabriz, Tabriz, Iran

Sep 2013 - Aug 2017

Advisor: Faramarz Talati

Department of Mechanical Engineering

Thesis title: Numerical Study of Mixed Convection in a Lid-Driven Cavity

Committee: Sima Baheri Islami

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher, Montreal, Canada

May 2025 - present

- Jet mixing and turbulent non-Newtonian flows, in collaboration with *SINTEF*
- Thermal convection in Micro-encapsulated PCM suspensions, in collaboration with *Universit de Lorraine*

PEER-REVIEWED PUBLICATIONS

M.M. Nasiri, **M.R. Daneshvar Garmroodi** and M. Tembely. Experimental study on elasto-capillary dynamics of a viscoelastic hallow droplet, *Under Preparation*.

M.R. Daneshvar Garmroodi and I. Karimfazli. Two-dimensional mixing stirred system, mixing regimes and transitions. *Under Preparation*.

M.R. Daneshvar Garmroodi and I. Karimfazli. Yield-stress fluid mixing: localisation mechanisms and regime transitions. *Journal of Fluid Mechanics*. 1021:A42, 2025.

M.R. Daneshvar Garmroodi and I. Karimfazli. Mixing in heterogeneous fluids: An examination of fluid property variations. *Journal of Non-Newtonian Fluid Mechanics*, 325, p.105196, 2024

M.R. Daneshvar Garmroodi and A. Ahmadpour. Numerical simulation of stratified waxy crude oil and water flows across horizontal pipes in the presence of wall heating. *Journal of Petroleum Science and Engineering*, 193:107458, 2020

M.R. Daneshvar Garmroodi and A. Ahmadpour. A numerical study on two-phase core-annular flows of waxy crude oil/water in inclined pipes. *Chemical Engineering Research and Design*, 159:362–376, 2020.

M.R. Daneshvar Garmroodi, A. Ahmadpour, and F. Talati. MHD mixed convection of nanofluids in the presence of multiple rotating cylinders in different configurations: a two-phase numerical study. *International Journal of Mechanical Sciences*, 150:247–264, 2019

M.R. Daneshvar Garmroodi, A. Ahmadpour, M. R. Hajmohammadi, and S. Gholamrezaie. Natural convection of a non-Newtonian ferrofluid in a porous elliptical enclosure in the presence of a non-uniform magnetic field. *Journal of Thermal Analysis and Calorimetry*, 1-17, 2019

CONTRIBUTED TALKS

M.R. Daneshvar Garmroodi and I. Karimfazli, Mixing Yield-Stress Fluids: Localization Mechanisms and Regime Transitions, *Second European Fluid Dynamics Conference*, Dublin, Ireland, August 2025.

M.R. Daneshvar Garmroodi and I. Karimfazli, Fluid mechanics of mixing in a two-dimensional stirred: regimes and mechanisms, *The Canadian Society for Mechanical Engineering International Congress*, Montreal, Canada, May 2025.

M.R. Daneshvar Garmroodi and I. Karimfazli, Scalar mixing in viscoplastic fluids, *26th International Conference of the Theoretical and Applied Mechanics*, Daegu, Republic of Korea, August 2024.

M.R. Daneshvar Garmroodi and I. Karimfazli, Understanding Mixing Dynamics in Yield Stress Fluids, *The Canadian Society for Mechanical Engineering International Congress*, Toronto, Canada, May 2024.

M.R. Daneshvar Garmroodi and I. Karimfazli, Scalar mixing in viscoplastic fluids, *The annual European rheology conference*, Leeds, United Kingdom, April 2024.

M.R. Daneshvar Garmroodi and I. Karimfazli, Localization of stirring flows: the effect of the yield stress, *8th PACIFIC RIM conference on rheology*, Vancouver, Canada, May 2023.

M.R. Daneshvar Garmroodi and I. Karimfazli, Chaotic mixing of complex fluids: on the effects of viscosity ratio, *The Canadian Society for Mechanical Engineering International Congress*, Edmonton, Canada, May 2022.

M.R. Daneshvar Garmroodi and I. Karimfazli, Mixing of heterogeneous fluids: buoyancy effects, *The annual European rheology conference*, Seville, Spain, April 2022.

M.R. Daneshvar Garmroodi and I. Karimfazli, Mixing of complex fluids: on the effects of inhomogeneity. *The Canadian Society for Mechanical Engineering International Congress*, May 2021.

PEER REVIEW - I reviewed two articles for this journal

Journal of non-Newtonian Fluids Mechanics

HONORS & AWARDS

Graduate Fellowship for Ph.D. program in Mechanical Engineering, Concordia University. Fall 2019
Four years, Monetary value: 37,915 CAD

Graduate Fellowship for Ph.D. program in Marine Engineering, NTNU University. Fall 2019
three years, Monetary value 479,600 NOK (approximately 70,000 CAD per year), declined.

Graduate Fellowship for Ph.D. program in Mechanical Engineering, Ghent University. Fall 2019
three years, Monetary value 30,000 Euro per year, declined.

Ranked 1st Among 60 Mechanical Engineering Master students of Amirkabir University. Spring 2019
four years, Monetary value: 300,000,000 Rial (approximately 7,000 CAD per year), declined.

Accepted in the First Stage of the National Computer and Mathematical Olympiads. Fall 2010, 2011, 2012

TEACHING

Instructor, SuperKids, Montreal, Canada 2024-present
Teaching linear algebra and calculus to high school and collage students

Teaching assistant, Concordia University, Montreal, Canada 2019-2025
Applied Advanced Calculus, Partial Differential Equations, Ordinary Differencial Equations, Fluid Mechanics I, Fluid Mechanics II, Heat Transfer I, Heat Transfer II, Thermodynamics I

Teaching assistant, Amirkabir University of Technology, Tehran, Iran 2017-2019
Fluid Mechanics, Heat Transfer I, Non-Newtonian Fluid Mechanics, Continuum Mechanics

SELECTED PROJECTS

- Developing an eddy-viscosity model for turbulent flows of Herschel-Bulkley fluids in OpenFOAM based on Lovato's work.

During PhD Program

- Numerical simulation of the impact of a hallow viscoelastic droplet on a wetting surface using Basilisk.

During PhD Program

- Adding dynamic mesh and tracking passive particles in the twoLiquidMixingFoam solver in OpenFOAM

During PhD Program

- Developing a melting solver in OpenFOAM Based on phase field and enthalpy methods

During PhD Program

- Coding annular flow of a Newtonian fluid by finite volume SIMPLE method in MATLAB

Advance Computational Fluid Mechanics Course

- Coding mixed convection in a lid-driven cavity using streamfunction-vorticity by finite difference method

Advance Computational Fluid Mechanics Course

- A priory and A Posteriori tests using dynamic Smagorinsky, and Wong LES Models one the filtered Burgers equation by finite Difference Method

Large Eddy Simulation Course

- Coupling Level-Set method with VOF algorithm in OpenFOAM

Multiphase Flows Course